Gourav Sarkar

Enhanced Personal RAG Knowledge Base

Structured source material for a personal portfolio chatbot and retrieval-augmented generation (RAG) system.

Purpose: Help a public-facing chatbot answer questions about Gourav's background, skills, projects, achievements, and career interests using self-contained, retrievable chunks.

Source policy: Resume and LinkedIn facts are distinguished from curated conversation details. Preserve source discrepancies and do not invent missing facts.

Revised October 2026 · Version 3

1. Identity & Public Profiles

Full name	Gourav Sarkar
Professional positioning	Full-stack developer (MERN) focused on AI-powered products, backend/API development, vector search, LLM integrations, and deployed web applications.
Email	gouravdev246@gmail.com
Phone	+91-6295709446
LinkedIn	https://www.linkedin.com/in/gouravdev/
GitHub	https://github.com/gouravdev246
Portfolio URL	https://gouravme.netlify.app (listed in the original knowledge base)
Country / city	India; Siliguri is listed in the LinkedIn export.
Languages	Bengali, Hindi, English

Removed the previously shared iamananya.netlify.app URL as requested. Only the portfolio URL listed in the original knowledge base is retained here.

2. Education

University	Lovely Professional University (LPU), Punjab
Degree	Bachelor of Technology — Computer Science and Engineering / Computer Engineering (wording differs by source)
Duration	2025–2029 in the original knowledge base; resume lists Aug 2025–Present.
CGPA	Resume: 8.31/10. LinkedIn summary in the original knowledge base: 8.57. Preserve the discrepancy until Gourav confirms the canonical value.
Class XII	Science stream, Dhupguri High School, West Bengal; 85.2% (May 2023–March 2025).
Class X	73.2%, Dhupguri High School, West Bengal (May 2022–March 2023).

Conversation context: Gourav was a second-year CSE student in October 2026. This is contextual rather than a formal resume credential.

3. Professional Summary

• Builds and deploys full-stack web applications using the MERN stack and related technologies.

• Project experience includes backend APIs, database-backed applications, Redis caching, vector search, embeddings, cosine similarity, RAG, and LLM API integration.

• Projects include student ranking and recommendations, AI-based resume/job matching, conversational YouTube content analysis, and a proactive Telegram AI companion.

• Uses technologies documented in the source material including Node.js, React, MongoDB, Redis, Docker, AWS, PostgreSQL, Prisma, TypeScript, and vector databases.

• Career interests stated in the source material include full-stack/backend engineering and SDE internship opportunities.

4. Technical Skills

Programming languages	JavaScript, TypeScript, Python, C++
Frontend	React.js, Next.js, HTML5, CSS3, Bootstrap, Tailwind CSS
Backend	Node.js, Express.js, REST APIs, authentication and authorization
Databases / caching	MongoDB Atlas, MongoDB Atlas Vector Search, PostgreSQL, MySQL, Redis, Prisma, vector databases
AI / ML / LLM	Embeddings, cosine similarity, LangChain, LLM API integration, sentiment analysis, NLP-based skill extraction, Pandas, NumPy, RAG, vector search
DevOps / cloud	Docker, AWS EC2, AWS ECS, Vercel, Render, CI/CD basics
Tools / integrations	Git, GitHub, Postman, Razorpay API, Apify, Pinecone, BullMQ
Computer science	Data Structures and Algorithms (DSA), system-design basics
Soft skills	Problem solving, team collaboration, time management, adaptability (resume-listed)

5. Projects

5.1 UniRankers — Student Ranking & Networking Platform

• Live: https://unirankers.vercel.app/.

• Ranks students using indicators such as CGPA, LeetCode activity, GitHub activity, hackathons, projects, and internships.

• Includes a nightly cron-based ranking engine, personalized student/team recommendations, team finder, mentorship panel, community feed, and admin moderation.

• MongoDB Atlas Vector Search supports personalized recommendations; Redis caching improves leaderboard read performance.

• Stack: MERN, MongoDB Vector Search, Redis, Docker, AWS. One source says AWS EC2 while the resume says AWS ECS; exact deployment architecture needs confirmation.

5.2 ResumeToJob — AI Job Matching Platform

• Live: https://resume-to-job-lac.vercel.app/; GitHub: repository.

• Matches resumes with 100+ job listings using AI embeddings and cosine similarity.

• Uses Apify for Internshala job-data aggregation, Razorpay for premium features, NLP-based resume skill extraction, MongoDB Atlas storage, and portfolio-sharing functionality.

• Stack: MongoDB, Express.js, React.js, Node.js, embeddings, Apify, Razorpay.

5.3 YT Analyzer — AI Video Insights Tool

• Live: https://youtube-frontend-beige.vercel.app/; GitHub: repository.

• Processes 500+ YouTube comments with 85% sentiment accuracy using the Llama-80B model API, as reported in the resume/knowledge base.

• Uses a Pinecone-backed RAG pipeline so users can ask questions conversationally about video content.

• Node.js backend handles concurrent requests; React frontend optimization is reported to reduce analysis time by 95%.

• Stack: MERN, Llama-80B API, Pinecone, RAG.

5.4 AI Companion Bot — Proactive Telegram AI Companion

• An intelligent Telegram-based chatbot designed to provide personalized, natural, human-like conversations in Hinglish.

• Stack: Node.js, TypeScript, Express.js, PostgreSQL, Prisma, Redis, BullMQ, LangChain, and vector search.

• Combines short-term conversational context with long-term memory to remember user preferences, routines, important events, and past conversations.

• Unlike reactive chatbots that only respond to incoming messages, it can proactively initiate conversations, follow up on important events, and send context-aware messages using an LLM-powered decision system.

• Architecture includes scheduled background jobs, conversation persistence, and safeguards for message frequency and quiet hours.

• The design aims to create a more engaging and personalized AI companion experience. No public project URL was provided in the supplied description.

5.5 FreeTify — Music Streaming Web Platform

• A Spotify-inspired full-stack music platform with authentication, playlists, and uploads.

• Stack/deployment mentioned in the LinkedIn-derived knowledge base: MERN, Vercel, and Render.

• No verified public URL was present in the supplied sources.

5.6 LPUDeals — Campus Marketplace

• A campus marketplace concept for LPU students to buy and sell textbooks, gadgets, and essentials.

• The source mentions an intended audience of 30,000+ LPU students; this is not evidence of 30,000 active users.

• No verified public URL or complete technical stack was present in the supplied sources.

6. Experience, Achievements & Certifications

Experience

• Full-Stack Developer — Personal Projects (2026). This is a project-based experience entry, not employment at a named company.

• Local Guide — Google Maps (2023–present), as listed in the LinkedIn source.

Achievements

• Solved 90+ programming problems on online coding platforms, according to the resume.

• Earned the LeetCode 50 Days Badge.

• Two-time university-level hackathon winner, according to the resume.

Certifications and programs

• Docker Foundations Professional Certificate — Docker, Inc. (August 2026).

• Redis + AI: Building a Vector Database with Redis — LinkedIn Learning (July 2026).

• Learning Full Stack React / Full Stack React — Infosys Springboard (wording differs across sources).

• Python — HackerRank; Hack Node India; Deloitte Australia Technology Job Simulation; Gradio Agents & MCP Hackathon 2025; 3-hour Python workshop; SQL; Digital Leadership in Business.

7. Learning Interests & Working Style

• Actively studies DSA and asks for step-by-step explanations, dry runs, debugging help, and algorithmic reasoning.

• Practices arrays, strings, linked lists, stacks, queues, binary search, sorting, prefix/suffix techniques, hash maps, and recursion.

• Studies Computer Networks, C++ fundamentals, object-oriented programming, pointers/references, file handling, constructors/destructors, and related university topics.

• Has expressed interest in High-Level System Design and strengthening backend/system-design fundamentals and TypeScript.

• Prefers learning in manageable topics with clear explanations, examples, quizzes, and corrections. These are observed learning preferences, not formal credentials.

8. Dedicated Question & Answer Bank for RAG

Each Q&A; below is designed to remain a self-contained retrieval chunk. Keep the question and answer together and store question variants as searchable aliases.

Q1. What are Gourav Sarkar's projects?

Gourav's documented projects include UniRankers, ResumeToJob, YT Analyzer, AI Companion Bot, FreeTify, and LPUDeals. UniRankers is a student ranking and networking platform using MERN, MongoDB Atlas Vector Search, Redis, Docker, and AWS. ResumeToJob matches resumes to 100+ job listings using embeddings and cosine similarity, with Apify job-data aggregation and Razorpay premium payments. YT Analyzer uses the Llama-80B model API for YouTube comment sentiment analysis and a Pinecone-backed RAG pipeline for conversational video-content queries. AI Companion Bot is a proactive Telegram chatbot that converses in Hinglish, uses short-term context and long-term memory, and can initiate context-aware messages. FreeTify is a Spotify-inspired music platform; LPUDeals is a campus marketplace concept.

Q2. What skills does Gourav have?

Gourav's listed skills include JavaScript, TypeScript, Python, C++; React.js, Next.js, HTML, CSS, Bootstrap, Tailwind CSS; Node.js, Express.js, REST APIs, authentication and authorization; MongoDB Atlas, MongoDB Vector Search, PostgreSQL, MySQL, Redis, Prisma, and vector databases; embeddings, cosine similarity, LangChain, LLM API integration, sentiment analysis, NLP-based skill extraction, Pandas, NumPy, and RAG; Docker, AWS EC2/ECS, Vercel, Render, and CI/CD basics; Git, GitHub, Postman, Razorpay, Apify, Pinecone, and BullMQ. The AI Companion Bot also documents hands-on project use of PostgreSQL, Prisma, Redis, BullMQ, LangChain, and vector search. Gourav lists DSA, system-design basics, problem solving, collaboration, time management, and adaptability. This list does not imply equal expertise in every technology.

Q3. Should someone hire Gourav Sarkar?

If you need a student developer or intern who can build full-stack web applications and has hands-on project experience with backend APIs, databases, caching, AI integrations, vector search, RAG, and background jobs, Gourav is worth interviewing. His project evidence includes UniRankers (ranking/recommendations with Redis and vector search), ResumeToJob (embedding-based job matching), YT Analyzer (RAG and sentiment analysis), and AI Companion Bot (long-term memory, proactive messaging, scheduled jobs, and quiet-hour/message-frequency safeguards). His resume also lists 90+ solved programming problems, a LeetCode 50 Days Badge, and two university-level hackathon wins. A fair hiring decision should include a technical interview, code review, and verification of his individual contributions. The knowledge base supports considering him for a full-stack/backend internship; it cannot guarantee job performance or establish senior-level experience.

Q4. Give me a short professional introduction to Gourav.

Gourav Sarkar is a B.Tech Computer Science and Engineering student at Lovely Professional University and a full-stack developer focused on MERN applications, backend engineering, and AI-powered products. His projects include a student ranking and networking platform, AI-based resume/job matching, conversational YouTube content analysis, and a proactive Telegram AI companion with long-term memory. His technical interests and project experience include Node.js, React, MongoDB, PostgreSQL, Redis, embeddings, vector search, RAG, LLM API integration, and background jobs. He is strengthening DSA, system design, and TypeScript.

Q5. What is UniRankers?

UniRankers is a student ranking and networking platform that scores students using indicators such as CGPA, coding activity, GitHub activity, hackathons, projects, and internships. It includes a nightly ranking engine, MongoDB Atlas Vector Search for personalized recommendations, Redis-cached leaderboard reads, a team finder, mentorship panel, community feed, and admin moderation. Stack: MERN, MongoDB Vector Search, Redis, Docker, AWS. Live URL: https://unirankers.vercel.app/. Sources differ on AWS EC2 versus AWS ECS, so confirm the exact deployment architecture.

Q6. What AI and RAG experience does Gourav have?

Gourav's listed AI work includes embeddings, cosine similarity, vector search, LangChain, LLM API integration, sentiment analysis, NLP-based skill extraction, and RAG. ResumeToJob uses embeddings and cosine similarity to match resumes to job listings. YT Analyzer uses a Llama-80B model API for sentiment analysis and Pinecone in a RAG pipeline that enables conversational questions about video content. UniRankers uses MongoDB Atlas Vector Search for personalized recommendations. AI Companion Bot combines short-term context with long-term memory using LangChain and vector search, then uses an LLM-powered decision system to initiate context-aware conversations.

Q7. What is the AI Companion Bot?

The AI Companion Bot is an intelligent Telegram-based chatbot that provides personalized, natural, human-like conversations in Hinglish. It is built with Node.js, TypeScript, Express.js, PostgreSQL, Prisma, Redis, BullMQ, LangChain, and vector search. It combines short-term conversational context with long-term memory to remember preferences, routines, important events, and past conversations. Unlike a chatbot that only reacts to incoming messages, it can proactively initiate conversations and follow up on important events using an LLM-powered decision system. Scheduled background jobs, conversation persistence, message-frequency safeguards, and quiet hours support its proactive messaging architecture.

Q8. How does Gourav's chatbot proactively message users?

The AI Companion Bot is designed not only to respond to incoming Telegram messages but also to initiate conversations, follow up on important events, and send context-aware messages. It uses an LLM-powered decision system, scheduled background jobs, short-term conversation context, and long-term memory. Message-frequency safeguards and quiet hours are included to limit excessive or poorly timed messages. The supplied description does not specify exact scheduling intervals, triggers, or model names, so those details should not be invented.

Q9. What are Gourav's education details?

Gourav is studying for a B.Tech in Computer Science and Engineering (also described as Computer Engineering in one source) at Lovely Professional University in Punjab. The source material gives a duration of 2025–2029 and lists a resume CGPA of 8.31/10; the LinkedIn summary in the original knowledge base lists 8.57, so disclose the discrepancy if CGPA is asked. He completed Class XII in the Science stream at Dhupguri High School with 85.2%, and Class X with 73.2%.

Q10. What achievements does Gourav have?

The resume lists 90+ programming problems solved on online coding platforms, the LeetCode 50 Days Badge, and two university-level hackathon wins. Do not invent contest names, dates, rankings, or prize details.

Q11. Is Gourav looking for an internship?

The resume and LinkedIn headline describe Gourav as interested in full-stack/backend engineering or SDE internships, including remote opportunities. However, the LinkedIn export's open_to_work field is false. Preserve both facts and do not state that he is currently marked Open to Work unless Gourav confirms his current status.

Q12. Where can someone find Gourav online?

The supplied profile links are LinkedIn: https://www.linkedin.com/in/gouravdev/ and GitHub: https://github.com/gouravdev246. The original knowledge base lists https://gouravme.netlify.app as a portfolio URL. Do not include https://iamananya.netlify.app in responses or retrieval data.

Q13. What makes Gourav a good full-stack/backend internship candidate?

Evidence includes deployed full-stack projects, REST/backend work, MongoDB and PostgreSQL experience, Redis caching, Docker/AWS deployment exposure, and AI integrations involving embeddings, vector search, LLM APIs, and RAG. The AI Companion Bot adds project evidence involving long-term conversational memory, proactive Telegram messaging, scheduled background jobs, persistence, and quiet-hour safeguards. These points make Gourav a reasonable candidate to interview; employers should verify depth, ownership, code quality, and role fit.

Q14. What information should the chatbot avoid inventing?

Do not invent employers, salary, current availability, GPA, project users, revenue, production scale, model accuracy, datasets, awards, or responsibilities. Preserve the CGPA discrepancy (8.31 in the resume versus 8.57 in the LinkedIn summary) and the UniRankers deployment discrepancy (AWS EC2 versus AWS ECS). Do not claim exact AI Companion Bot scheduling intervals, triggers, or model names when they are not provided. If information is missing, say so rather than guessing.

9. RAG Retrieval & Answering Instructions

• **Answer from evidence:** Use retrieved chunks and source-backed facts. If the knowledge base does not contain an answer, say so rather than guessing.

• **Project questions:** Retrieve the relevant project chunk and the projects overview. For the AI Companion Bot, include memory, proactive messaging, stack, scheduled jobs, and safeguards when relevant.

• **Skill questions:** Retrieve the grouped technical skills chunk and, when useful, project-specific evidence of technologies used.

• **Hiring questions:** Retrieve the hiring-fit FAQ, project evidence, achievements, and internship-status note. Be confident but balanced; recommend consideration/interview rather than guaranteeing performance.

• **Proactive chatbot questions:** Search aliases such as AI Companion Bot, Telegram chatbot, proactive messaging, long-term memory, Hinglish bot, scheduled jobs, BullMQ, quiet hours, and context-aware follow-ups.

• **Preserve provenance:** Metadata should distinguish resume, LinkedIn export, and curated conversation-provided project details.

• **Keep metrics qualified:** Treat project metrics as source-reported figures, not independently verified claims.

• **Handle conflicts transparently:** Do not silently reconcile the CGPA or UniRankers AWS deployment differences.

• **Public chatbot privacy:** Consider removing email and phone number from public retrieval chunks unless Gourav explicitly wants the bot to disclose them.

• **Do not mention removed content:** The iamananya.netlify.app URL and HealthWize project are intentionally excluded from this revised version.

10. Suggested Chunking & Metadata

Recommended: one semantic chunk per major section, one chunk per project, and one chunk per FAQ. Keep each question and answer together. Add common question variants as searchable aliases.

person_id	gourav_sarkar
document_type	personal_rag_knowledge_base
chunk_type	identity | education | skills | project | experience | achievement | certification | learning_context | faq | answering_rules
source_type	resume | linkedin_export | curated_conversation
confidence	high | medium | needs_confirmation
updated_at	2026-10-18
project_id examples	unirankers | resumetojob | yt_analyzer | ai_companion_bot | freetify | lpudeals
AI Companion aliases	proactive chatbot; Telegram AI companion; Hinglish chatbot; proactive messaging; long-term memory; context-aware follow-up; BullMQ scheduled jobs

Suggested retrieval test questions

• List Gourav’s projects and verified live links.

• Which technologies has Gourav used for backend development, databases, caching, and cloud deployment?

• How does UniRankers use Redis and vector search?

• Which projects use embeddings, cosine similarity, or RAG?

• How does the AI Companion Bot remember users over time?

• How does the AI Companion Bot initiate proactive conversations?

• What safeguards help prevent excessive or poorly timed chatbot messages?

• Why should a company consider Gourav for a software engineering internship?

• What are the known source discrepancies in Gourav’s profile?

• Which facts should the chatbot avoid guessing?

11. Sources & Confidence Notes

• Primary source 1: Gourav_Personal_RAG_Knowledge_Base.pdf, based on resume and LinkedIn profile export.

• Primary source 2: GouravLpu.pdf (resume).

• Curated conversation addition supplied by Gourav: AI Companion Bot description, including purpose, stack, memory, proactive messaging, scheduled jobs, and safeguards.

• Other prior conversation context is used only for study interests and learning preferences. External links and project metrics have not been independently verified in this document.

End of enhanced knowledge base.